21 September, 2025

- **Compiler**
  - Uses mathematical models - Abstract ones.
  - Generates machine code.
  - Parts
    - Syntax analyzer
    - Semantic analyzer
    - Lexical analyzer
- **Basics of English**
  - Characters - Fundamental blocks
  - Words / Strings - Combination of characters
  - Sentence - Combination of words
    We need to check for validity of strings and sentences
- **Language**
  - A certain specified set of strings of characters from alphabets (a valid 'set' of characters).
- Some rules
  - a^0 = Λ (Null String). Similar to phi (empty) set in mathematics.
  - a.Λ = a
  - Λ.Λ = Λ
  - No multiplication. Instead concatenation (addition of strings in code). Not commutative. Order matters (unlike numerical multiplication)
- **Operators**
  - Kleene Star (`*`) - Makes an infinite language
    - If I have a set like `L = [a, b]`, then `L*` would be any combination of all the characters of `L`. `L* = [Λ, a, b, aa, bb, aab, aaa, ...]`. Basically all combination of letters. (The Λ is a combination of $a^{0}.b^{0}$)
    - Lexicographic order: Lengthwise combination in alphabetic order. The lexicographic combination of the set `L` will look like `[Λ, a, b, aa, ab, ba, bb, aaa, aab, aba, abb, baa, bab, ...]`.
    - If we have `L = [a, b]` and `M = [b, a]`, `L* = M*`.
  - Plus (`+`) - Makes an infinite language without Λ.
    - If I have a set like `L = [a, b]`, then `L+` is same as `L*` without `Λ`. So, it would be `[a, b, aa, ab, ba, bb, ...]`.

        - If the character set has empty character, it would eventually be yielded. For example, if I have `L = [Λ, a, b]`, `L+ = [Λ, a, b, aa, ab, ...] = L*`.

- **Functions**
    - Length function
        - Returns length of a string. Count of characters.
        - Reverse function
            - Returns reversed order of the a string.
        - Palindrome `[Λ plus all strings X, such that Reverse(X) = X]`.
            - `L = [a, b]`. `Palindrome(L) = [Λ, a, b, aa, bb, ...]`.
              Introduction to Computer Theory - Daniel I. A Cohen